AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 10 · STUDENT LESSON

Baselines First

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit the reference. Freeze predictions. Compare the same rows.

Judge supplied improvements against training-derived and declared transparent baselines.

Keep one habit: Training majority → validation trace → leaf contrast → complete two-baseline lab → independent fresh labels.

MAIN LESSON · SECTION 1

What does a simple reference method achieve?

> An accuracy percentage needs a task, a denominator and a fair comparison.

Learning goals

I can fit a **majority label baseline** using **training** labels only.

I can score fixed **prediction**s on the same validation examples.

I can explain what a **baseline** comparison does and does not establish.

Recall source-separated evaluation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.
- Why should frames from the same trip stay in one set when evaluating on new trips?

End goal

By the end, I can explain **baseline**s first using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A model must be compared with a simple transparent strategy on the same untouched examples.



LEARNING TARGET A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Choose the constant from training labels only

> A simple baseline can still fit a value from training evidence.

Purpose and new fictional source

A learner reports that a classifier gets 80 percent correct. Is that useful? We need to know the task, the examples and what a simple method achieves on those same examples. A **baseline** is a reference method.

This week our **baseline** ignores each scene and always predicts the label that is most common in the **training** set.

Use a new teacher-authored fictional dataset for predicting clear or blocked lanes. Each listed row describes one frame from a different trip on a familiar route; **training** and validation trips do not overlap. This is a new practice set, not the twelve-frame table from Week 8. All supplied labels have been resolved under the classroom rule. There are no review cases in this exercise, and no real navigation claim is made.

Complete training/validation labels

Set	Labels in row order	Use
Training	clear clear blocked clear blocked clear	Choose the constant label
Validation	blocked clear blocked clear	Score the fixed predictions
Final test	Not supplied in this exercise	Keep for later final evaluation

Majority counts and declared tie rule

The **training** counts are four clear and two blocked. Our stated method therefore predicts clear for every new example. A **baseline** can still learn a value from **training** data; simple does not mean it uses no data.

If the two **training** counts tie, our rule chooses clear, declared before scoring.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **baseline**s first to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Fit, freeze and score the majority baseline

> Keep training majority share separate from validation accuracy.

Complete supplied majority script

```
train = ["clear", "clear", "blocked",
         "clear", "blocked", "clear"]
valid = ["blocked", "clear", "blocked", "clear"]
prediction = "clear"
if train.count("blocked") > train.count("clear"):
    prediction = "blocked"
correct = 0
for label in valid:
    if prediction == label:
        correct = correct + 1
print(prediction, correct, len(valid))
```

Trace clear2/4 on validation

The **output** is clear 2 4. The fixed **prediction** matches two of the four validation labels, so **accuracy** is 2/4 or 50 percent. The **training** majority share, 4/6, is not the validation **accuracy**.

The **baseline** does not inspect the validation labels to decide which constant to predict.

Keep fitting separate from scoring

EVIDENCE	OPERATION	RESULT
Training labels only	Count clear and blocked	Store prediction clear
New validation examples	Repeat stored prediction	clear for every row
Validation labels	Compare after prediction	2 correct out of 4

The baseline uses training labels to choose its constant. Validation labels are used to score that fixed constant, not to refit it.

State empty-data and allowed-label assumptions

The code assumes nonempty **training** data with only clear and blocked labels. It counts matches on the supplied validation list; a percentage also requires a nonempty validation set.

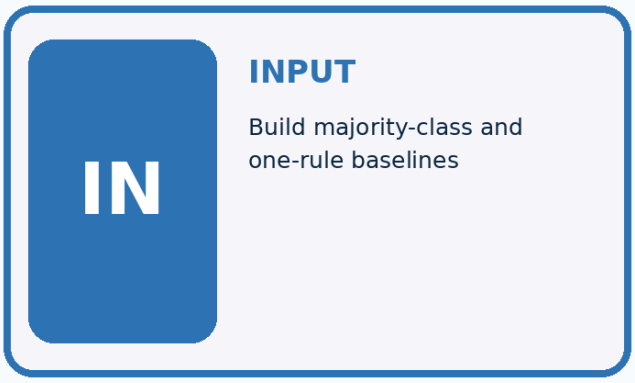
Do not run it on empty **training** data and describe the default clear value as learned **evidence**. Other labels and unresolved review statuses need an explicit policy.

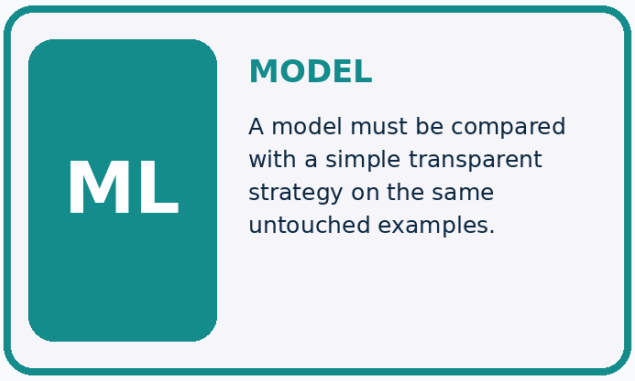
▼ **Earlier wording, alternatives and picture transcripts**

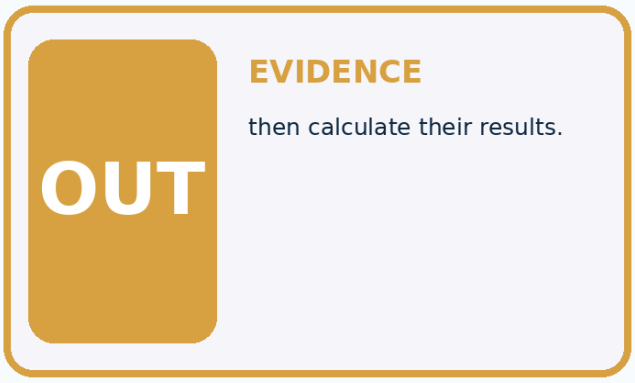
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Build majority-class and one-**rule baseline**s, then calculate their results.

MODEL IDEA A model must be compared with a simple transparent strategy on the same untouched examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Test set has 12 healthy and 8 spotted leaves. Always-healthy **baseline** gets 12 ÷ 20 = **60% accuracy** but 0% spotted recall.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

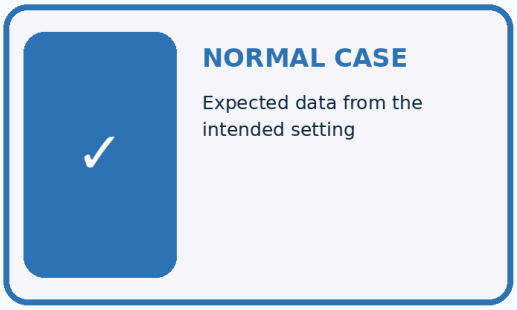
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

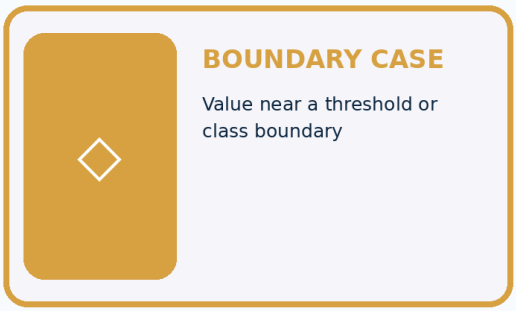
Compare overall correctness with the errors it hides

> Use identical rows and label rules for every method.

Four evaluation conditions



NORMAL Expected data and expected behavior.

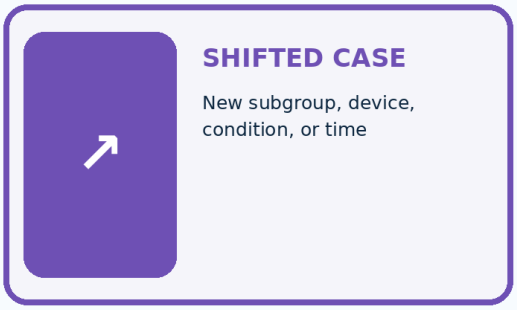


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Change the third candidate **prediction** from spotted to healthy.

▼ **Check after predicting**

Prediction: Candidate becomes correct on all 4 validation cases.

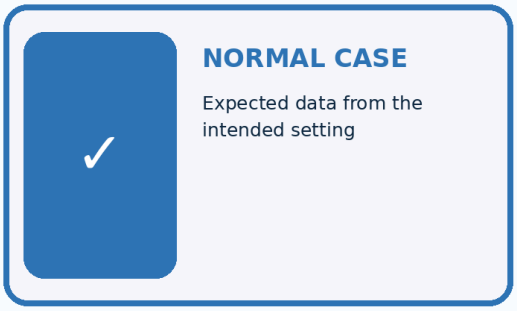
Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

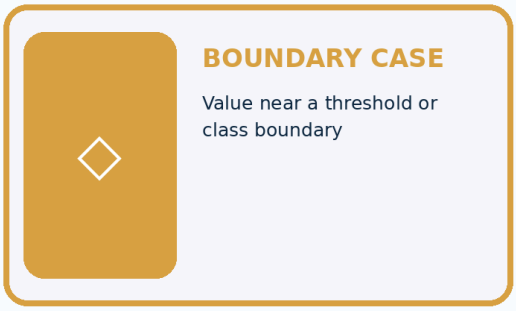
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

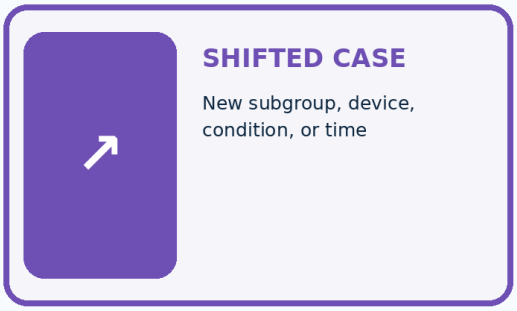
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace matched rows before comparing headline percentages

> An extra correct case is visible evidence; future performance remains unknown.

Separate pictured leaf facts and comparison scope

Make a simple **baseline** earn its place before claiming a better classifier.

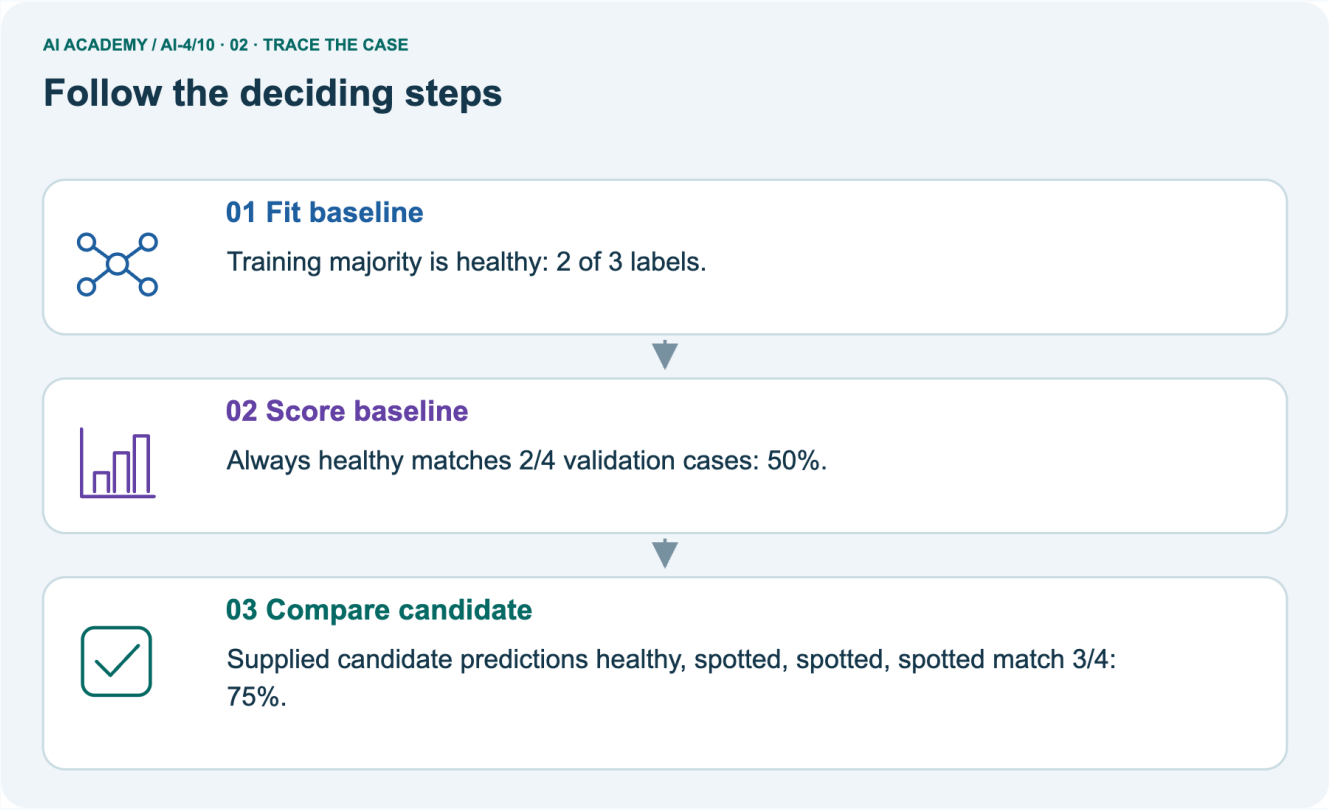
Training labels: healthy, healthy, spotted. Majority **baseline** predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured leaf example and corridor example both start at2/4 and compare a3/4 candidate, but their labels and wrong-row identities differ. Corridor baseline missesV1,V3; supplied candidate missesV3. Leaf baseline missesrows2,4; leaf candidate missesrow3.

Fixing only the leaf candidate's third prediction gives4/4 on that same development set; it is an answer-informed illustrative repair, not new independent validation. The original50%→75% difference is25percentage points; relative gain in accuracy is25/50=50%, a different quantity.

Trace majority, candidate and denominators



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit baseline	Training majority is healthy: 2 of 3 labels.
2. Score baseline	Always healthy matches 2/4 validation cases: 50%.
3. Compare candidate	Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

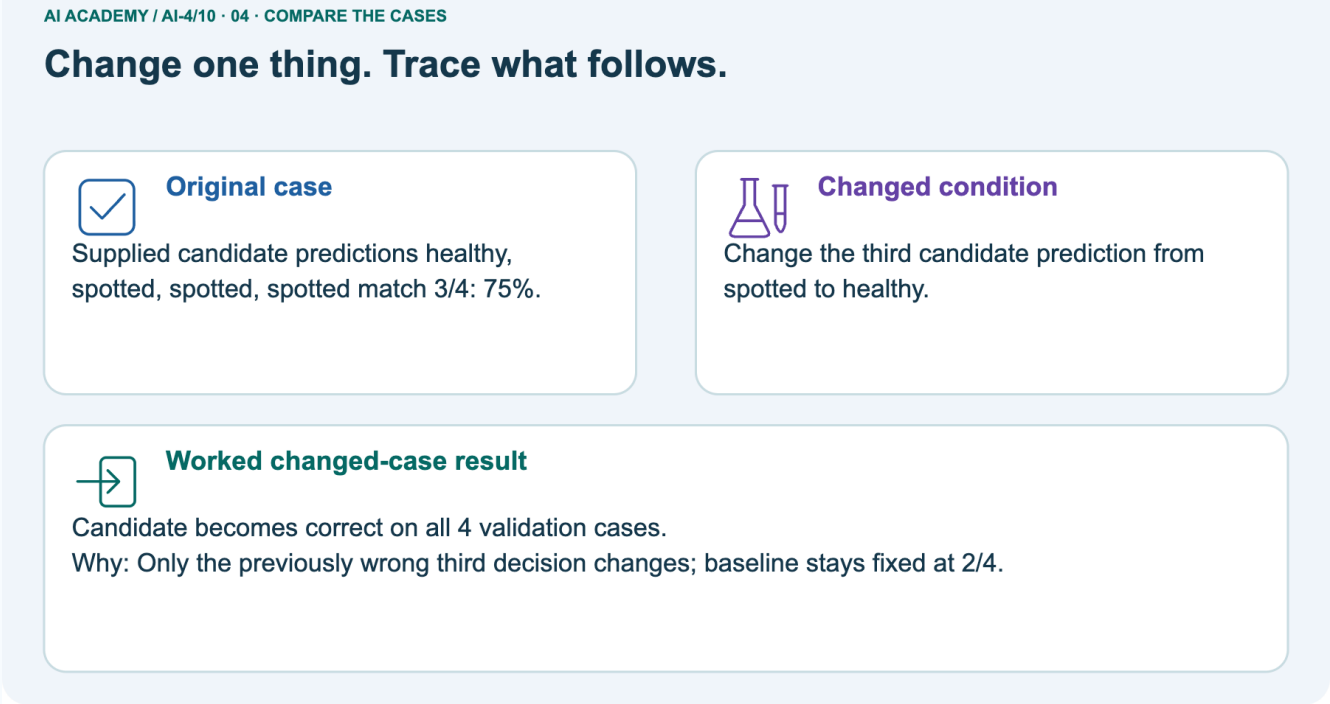
Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict fixing the pictured third decision

- Change one thing: Change the third candidate **prediction** from spotted to healthy. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain evaluation limits

- These supplied **prediction**s illustrate comparison. No actual model was trained, and four cases are limited **evidence**.

- Read the labelled picture: Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the corridor supplied comparison

Suppose a candidate method supplies blocked, clear, clear, clear for the four validation rows in the table. These are supplied **prediction**s for a comparison exercise; we have not fitted that candidate.

It gets three of four correct, or 75 percent. The **baseline** gets two of four, or 50 percent. The candidate is ahead by one correct example and 25 **percentage point**s on this set.

Use identical row IDs, reference labels, exclusions and metric for both methods. Match **prediction**s to their rows even if files are reordered. A score on easier examples is not a fair comparison.

Neither this four-row result nor beating one **baseline** proves that the candidate will perform well on future trips.

> **The candidate misses the third validation row. Which rows does the baseline miss? Explain how the candidate changes the number correct. Why is the improvement 25 percentage points rather than a claim of 25 percent relative improvement?**

Record **training** counts, the fixed **prediction** and tie rule, validation IDs and labels, **prediction** lists and correct/total counts. Keep the final test separate from development decisions.

Next week we will search candidate rules using **training evidence**, then compare choices carefully.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 10: Baselines First

STEP	EVIDENCE AND REASONING
1	Test set has 12 healthy and 8 spotted leaves.
2	Always-healthy baseline gets 12 ÷ 20 = 60% accuracy but 0% spotted recall.
3	On the same 20 leaves, the visible-spot rule correctly labels 5 spotted leaves, misses 3 spotted leaves, correctly labels 10 healthy leaves and falsely marks 2 healthy leaves as spotted. Accuracy=(5+10)/20=75%; spotted recall=5/8=62.5%.
4	A learned model must beat relevant baseline metrics—not only one convenient total.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 Test set has 12 healthy and 8 spotted leaves.

STEP 2 Always-healthy **baseline** gets 12 ÷ 20 = **60% accuracy** but 0% spotted recall.

STEP 3 On the same 20 leaves, the visible-spot rule correctly labels 5 spotted leaves, misses 3 spotted leaves, correctly labels 10 healthy leaves and falsely marks 2 healthy leaves as spotted.

Accuracy=(5+10)/20=75%; spotted recall=5/8=62.5%.

STEP 4 A **learned model** must beat relevant **baseline** metrics—not only one convenient total.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Baselines First

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find equal accuracies hiding different mistakes

> A high majority accuracy can miss every example of a rarer label.

Complete90% example and limits

- Imagine ten validation cases with nine clear and one blocked, while the **training**-derived constant is clear. The **baseline** gets 9/10 or 90 percent **accuracy** but misses the only blocked case.

  Overall **accuracy** counts all rows equally; it does not describe every kind of error or its consequences. Later weeks will examine those errors in detail.

- A majority **baseline** is a useful first comparison, not a guarantee of the best simple method. It can be weak when validation label proportions differ from **training**.

  Keep the declared **baseline**, report what happened and consider other justified reference methods during development. Do not tune a final test result into a better-looking claim.

> **A candidate and the clear baseline both get 90 percent on ten cases. Does that prove they make the same mistakes or are suitable for use? State what additional evidence you would inspect.**

Challenge complexity as proof of quality

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Any trained model is more intelligent and therefore better than a simple rule.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Any trained model is more intelligent and therefore better than a simple rule. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise the unchanged baseline on the same validation rows

> Do not choose a new constant from evaluation labels.

Try the supplied W2

Use workbook W2 for the connected example “A **baseline** before a model”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Reject evaluation-driven constants and different-row comparisons

A learner sees that blocked is common in a new validation set and changes the **baseline** to always blocked. That is not the original **training**-majority **baseline**. Its parameter was selected from evaluation labels.

Restore the **training**-derived constant for that comparison and document any separate method honestly.

Another learner compares a candidate on ten easy rows with a **baseline** on four difficult rows. The difference could come from the examples.

Score both on the same intended set with the same treatment of missing **prediction**s and unresolved labels; report exclusions rather than silently dropping mistakes.

Check whether simple means no training evidence

- Use the worked case for A **baseline** before a model. Proposed explanation: “A simple **baseline** does not need **training** data”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured validation comparison

- Can the candidate’s **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
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Reserve the independent new labels

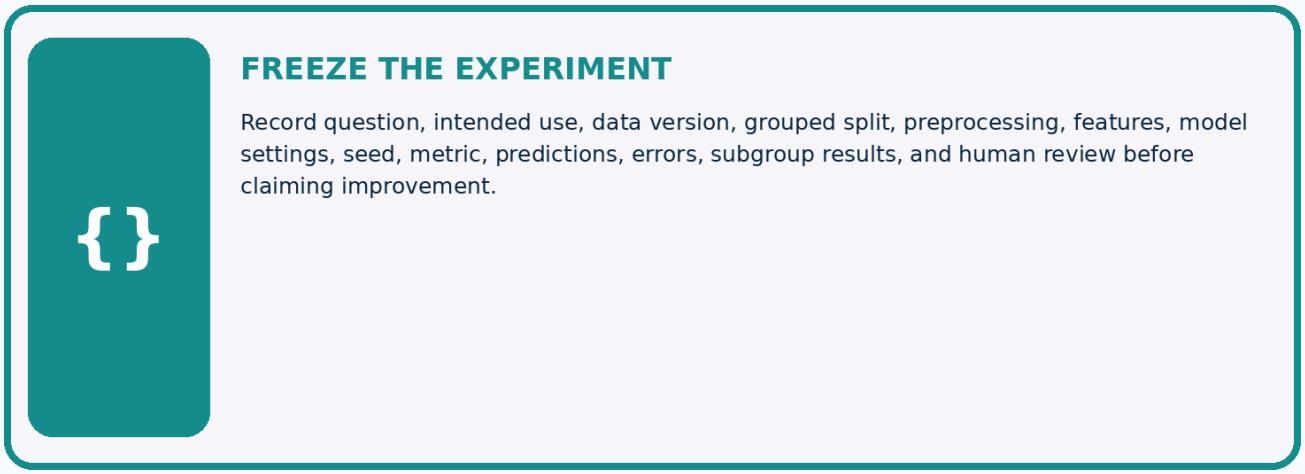
This is supported practice. Keep the separately named independent case for an attempt before feedback.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.
</details>



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Calculate the validation score

Check your reasoning

Explain the picture

Run two complete baselines on one fixed row set

> Compare both total correctness and missed blocked cases.

Complete row-aligned feature and reference table

Fixed validation row · Available marker count · Reference label · Training-majority prediction · Declared marker>=6 rule

> **Fixed validation row:** V1
> **Available marker count:** 7
> **Reference label:** blocked
> **Training-majority prediction:** clear
> **Declared marker>=6 rule:** blocked

> **Fixed validation row:** V2
> **Available marker count:** 2
> **Reference label:** clear
> **Training-majority prediction:** clear
> **Declared marker>=6 rule:** clear

> **Fixed validation row:** V3
> **Available marker count:** 4
> **Reference label:** blocked
> **Training-majority prediction:** clear
> **Declared marker>=6 rule:** clear

> **Fixed validation row:** V4
> **Available marker count:** 1
> **Reference label:** clear

> **Training-majority prediction:** clear
>
> **Declared marker>=6 rule:** clear

Declared methods and checked scope

For this two-baseline activity the feature is an invented marker count available before prediction. The teacher declares the transparent rule blocked when count>=6, otherwise clear, before the comparison. It is not searched or fitted using these validation labels. Both methods use V1–V4 and the same resolved references. Majority baseline:2/4correct, blocked found0/2.

Rule baseline:3/4correct, blocked found1/2; each gets both clear rows correct. These are toy methods on four invented rows; neither rule is real lane-safety guidance.

Normal, tie, empty and changed-proportion probes

Complete baseline probe	Expected bounded result
Normal original training/validation labels	Choose clear from4vs2 training counts; score2/4.
Boundary training clear,blocked	Tie policy chooses clear before validation scoring.
Missing empty training list	No learned majority; stop/request data rather than claim a default is evidence.
Changed validation labels blocked,blocked,blocked,clear; training unchanged	Keep clear constant:1/4correct, blocked found0/3. Do not retune the baseline from evaluation labels.

Optional complete guarded comparison program

```
def fit_majority(labels):
    if not labels or any(label not in {'clear', 'blocked'} for label in labels):
        raise ValueError('Need nonempty resolved clear/blocked training labels')
    # The tie policy is declared before looking at validation labels.
```

```
    return 'blocked' if labels.count('blocked') > labels.count('clear') else 'clear'


def score(references, predictions):
    if not references or len(references) != len(predictions):
        raise ValueError('Need nonempty, same-case paired labels and predictions')
    if any(label not in {'clear', 'blocked'} for label in references + predictions):
        raise ValueError('Unresolved or unsupported label')
    correct = sum(actual == predicted for actual, predicted in zip(references,
predictions))
    blocked_total = references.count('blocked')
    blocked_found = sum(actual == predicted == 'blocked'
                        for actual, predicted in zip(references, predictions))
    return {'correct': correct, 'total': len(references),
            'blocked_found': blocked_found, 'blocked_total': blocked_total,
            'blocked_recall': blocked_found / blocked_total if blocked_total else None}


training = ['clear', 'clear', 'blocked', 'clear', 'blocked', 'clear']
constant = fit_majority(training)
# V1–V4 share this fixed order, available feature and resolved reference label.
feature = [7, 2, 4, 1]
reference = ['blocked', 'clear', 'blocked', 'clear']
majority_predictions = [constant] * len(reference)
# Teacher-declared rule, fixed before this comparison; not fitted on validation.
rule_predictions = ['blocked' if value >= 6 else 'clear' for value in feature]
print(constant, majority_predictions, score(reference, majority_predictions))
print(rule_predictions, score(reference, rule_predictions))
```

Keep the evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Baseline Challenge: calculate two baselines on one fixed set, compare class-specific errors, and choose the honest benchmark.

1. Write the hypothesis and expected **evidence** before running.

2. **Baseline** Challenge: calculate two **baseline**s on one fixed set, compare class-specific errors, and choose the honest benchmark.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit and compare on fresh labels

> Show both row-level matches before reporting the percentage-point difference.

Complete independent task and scope

New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

Keep this task's new training labels and validation order separate from the worked corridor and pictured leaf cases. Use the same majority rule with the predeclared clear-on-tie policy. Candidate predictions are supplied; no candidate fitting or future-performance result is provided.

First attempt and support record

> **My independent explanation, trace and limits**
>
> | |
> |---|
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> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Fit and compare independently

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A dataset has 18 grass and 2 fern examples. A **baseline** always predicts grass. Compute **accuracy** and name a serious limitation.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Record majority and transparent-**rule baseline**s for the Plant Classifier using the frozen test plan.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the reference methods beside the project result

> A later learned model needs the same fair comparison contract.

Project artifact and record

THIS WEEK'S CHECKPOINT Record majority and transparent-**rule baseline**s for the Plant Classifier using the frozen test plan.

Record the exact dataset/label version, training counts and chosen constant, tie rule, transparent-rule definition and feature timing. Compare fixed predictions on the same validation IDs; report correct/total and label-specific errors. Retain exclusions and unresolved cases under an explicit policy.

Keep any final test separate from development choices; beating these baselines on a tiny set does not establish deployment readiness.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence and human limits

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

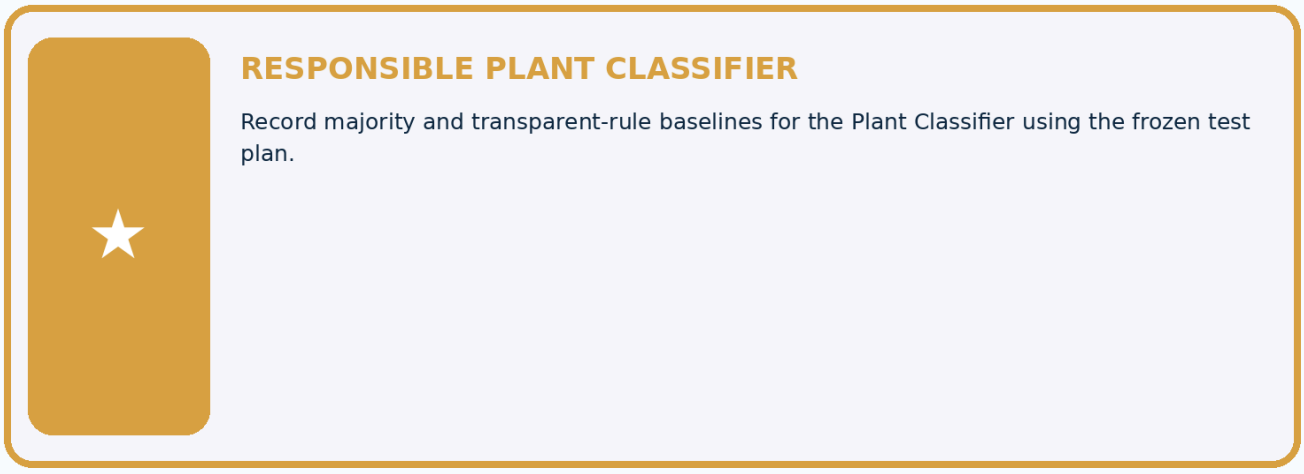
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a reference score and the error it misses

> Explain the evidence needed before a model can be called better.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Fit a **majority label baseline** using **training** labels only

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
baseline	simple comparison method	
majority class	most frequent training label	
rule baseline	transparent fixed decision rule	
incremental value	measured gain beyond baseline	

Closer-explanation terms

Word	Meaning
Baseline	A simple reference method used to judge a more complex approach
Majority label	The label with more examples than the other label in this binary task
Accuracy	Correct predictions divided by the number of scored examples

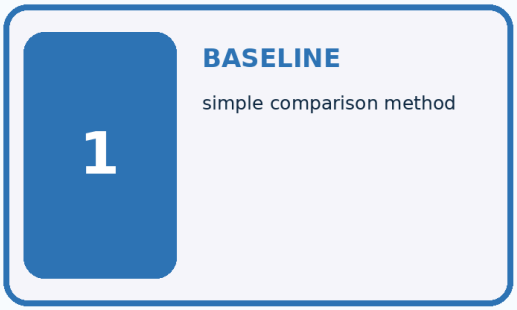
Word	Meaning
Percentage point	A unit for the difference between two percentages

▼ Earlier wording, alternatives and picture transcripts

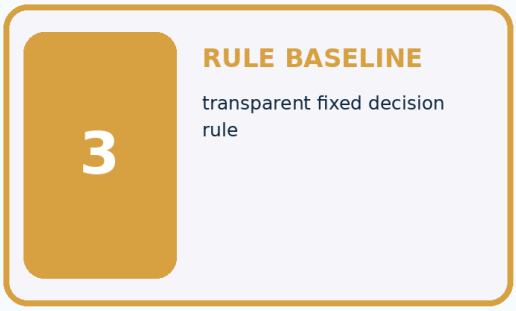
VOCABULARY

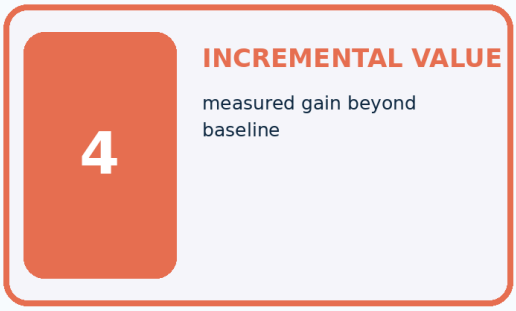
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Any trained model is more intelligent and therefore better than a simple rule.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **baseline**, **majority class**, **rule baseline**, **incremental value**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Any trained model is more intelligent and therefore better than a simple rule.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

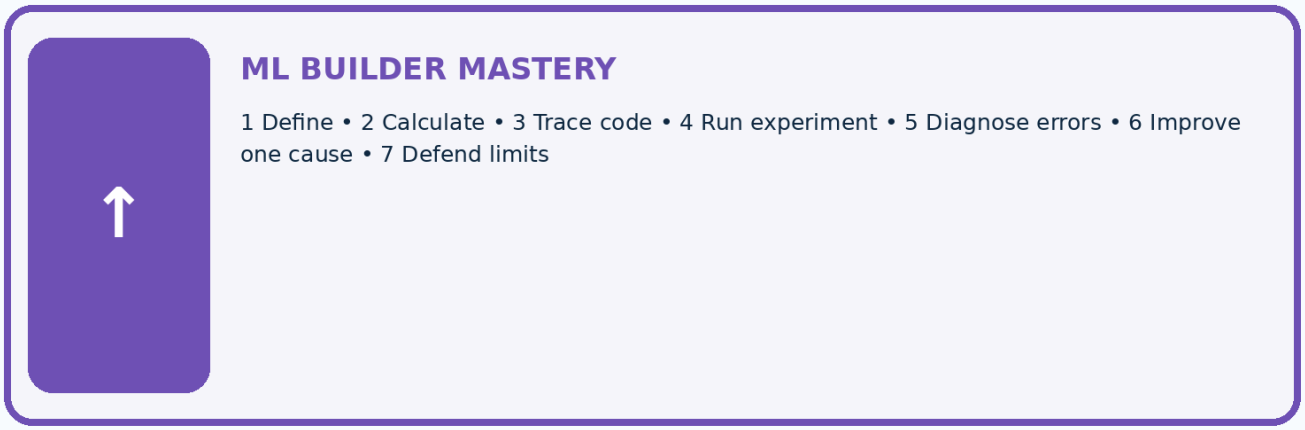
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the fresh majority-fitting and candidate comparison task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

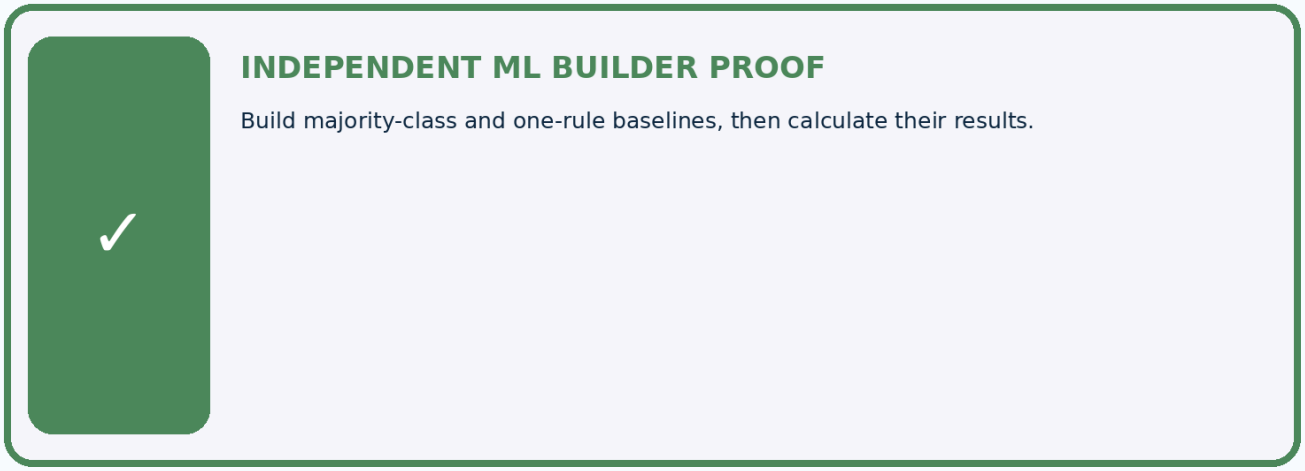
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Build majority-class and one-**rule baseline**s, then calculate their results.

NEXT BRIDGE Carry the experiment **evidence** into: Decision Trees.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.

Calculate and critique **accuracy**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

Evidence or stage	Value or result
Overall correct	18/20
Spotted found	0/2

▼ Earlier wording, alternatives and picture transcripts

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A majority baseline can miss every positive

A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.

Calculate and critique **accuracy**.

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Fit a majority label baseline using training labels only	Explain the deciding step, show a trace or test, and state one limit.
Score fixed predictions on the same validation examples	Explain the deciding step, show a trace or test, and state one limit.
Explain what a baseline comparison does and does not establish	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Make a simple **baseline** earn its place before claiming a better classifier.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Make a simple **baseline** earn its place before claiming a better classifier.

- Which labelled step matches this detail? Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.
- 1. Fit **baseline**
- 2. Score **baseline**
- 3. Compare candidate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare candidate Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Read the picture in order.

Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Change the third candidate **prediction** from spotted to healthy. Which conclusion is supported by the supplied facts?

- 1. Candidate becomes correct on all 4 validation cases.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Candidate becomes correct on all 4 validation cases. Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Repair and keep the record

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Define the population and conditions in an intended-use statement. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table schema. Short practice: Why should the missing value not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

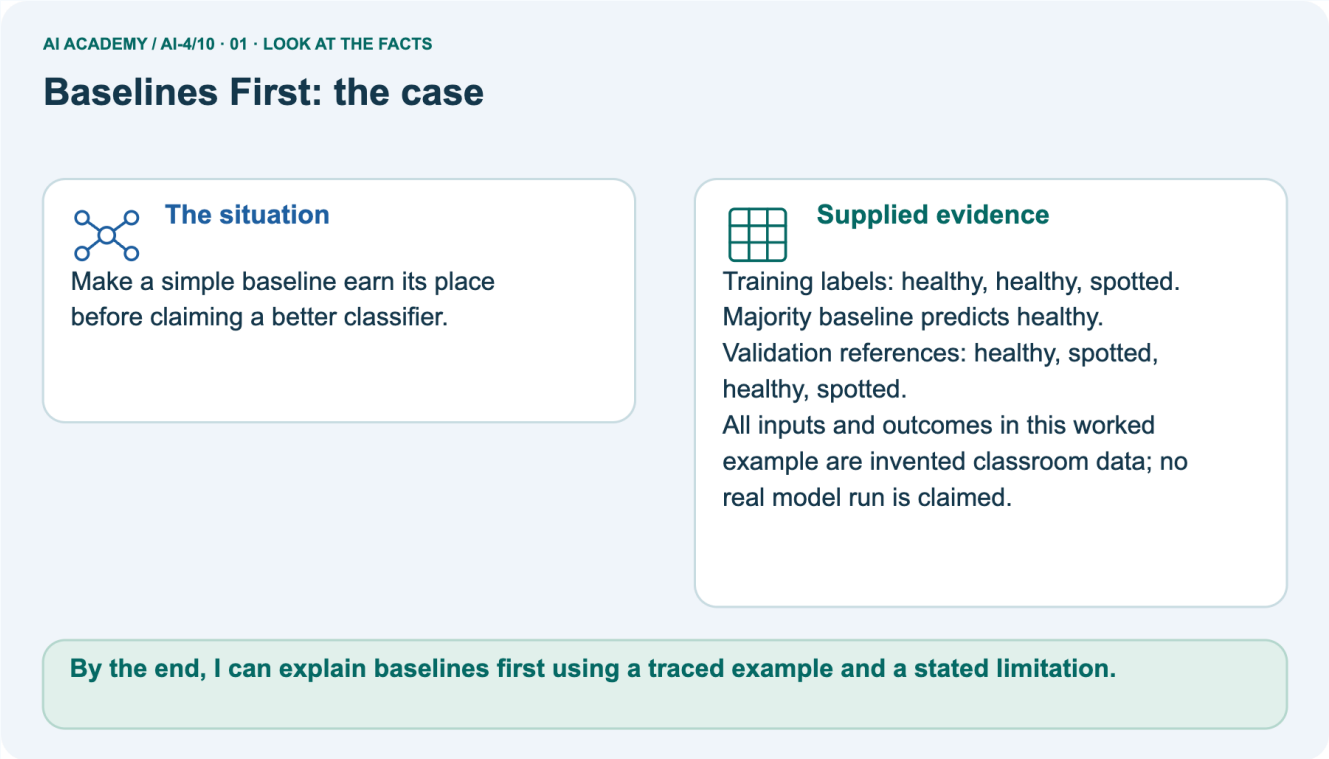
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Baselines First: the case



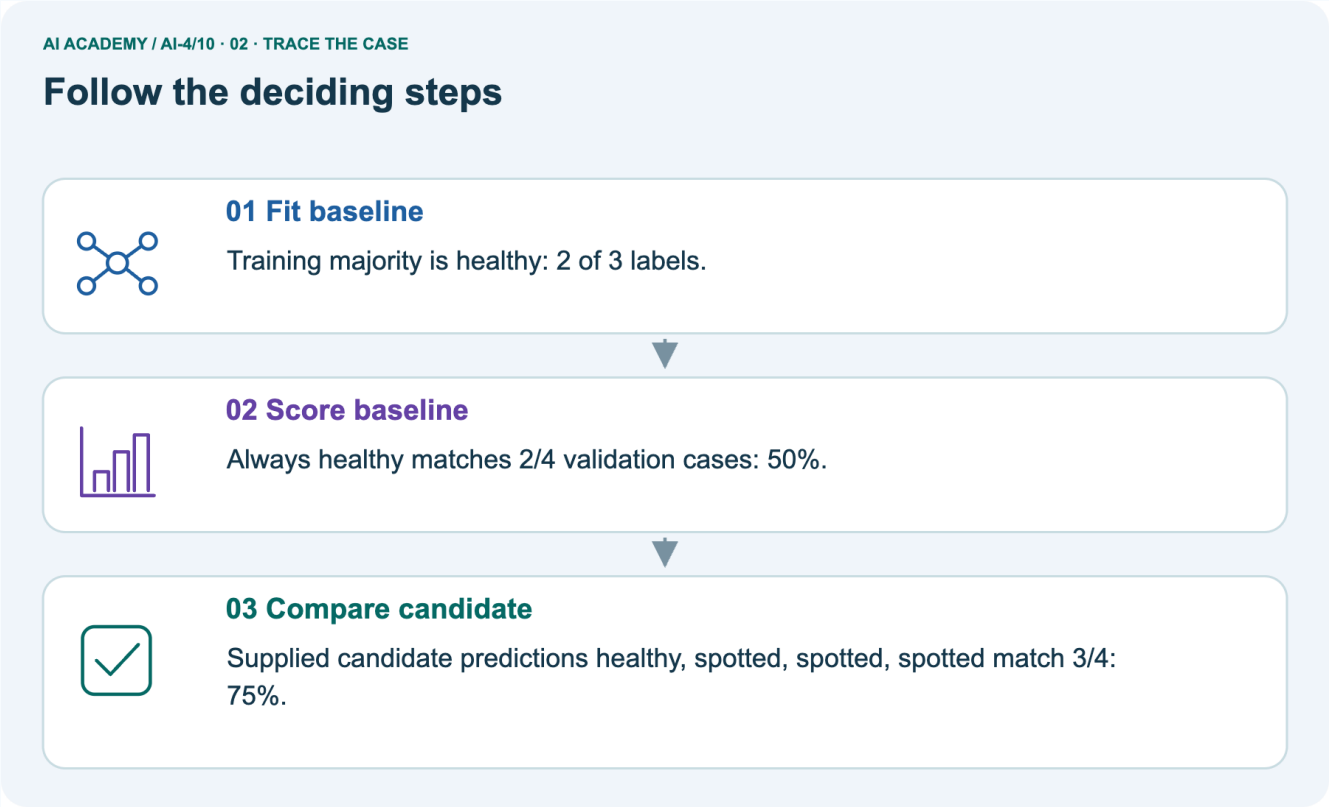
Follow the labelled picture with the matching explanation. Open the vector diagram

Make a simple **baseline** earn its place before claiming a better classifier. Supplied facts: **Training** labels: healthy, healthy, spotted. Majority **baseline** predicts healthy.

Validation references: healthy, spotted, healthy, spotted. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

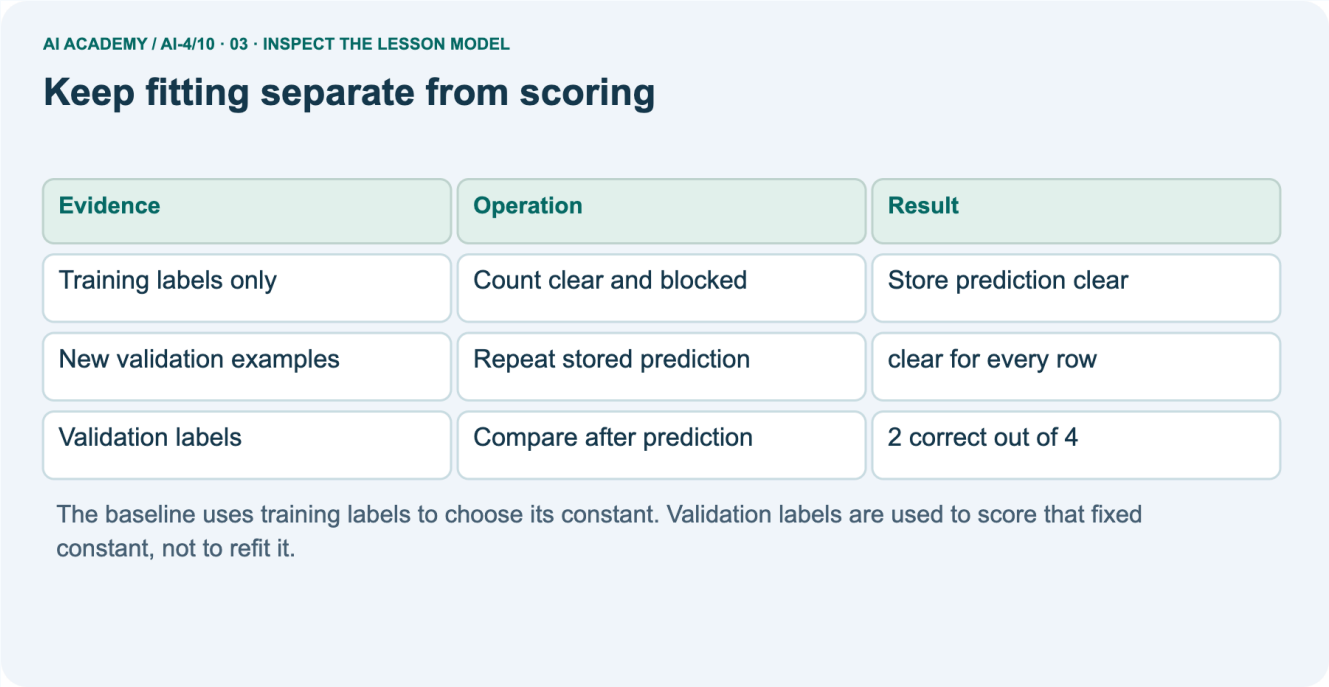


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **baseline**: **Training** majority is healthy: 2 of 3 labels. → Score **baseline**: Always healthy matches 2/4 validation cases: 50%. → Compare candidate: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep fitting separate from scoring



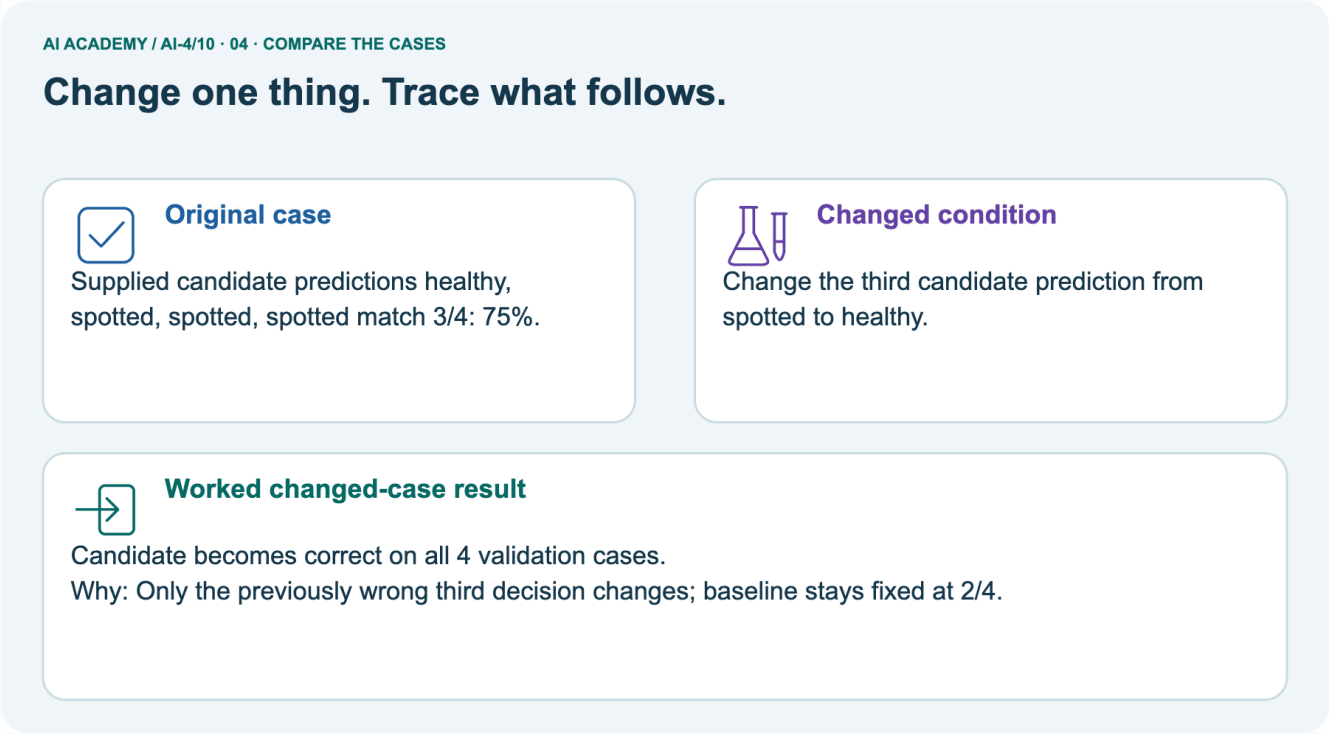
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **fitting** separate from scoring. **Training** labels only | Count clear and blocked | Store **prediction** clear. New validation examples | Repeat stored **prediction** | clear for every row.

Validation labels | Compare after **prediction** | 2 correct out of 4

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



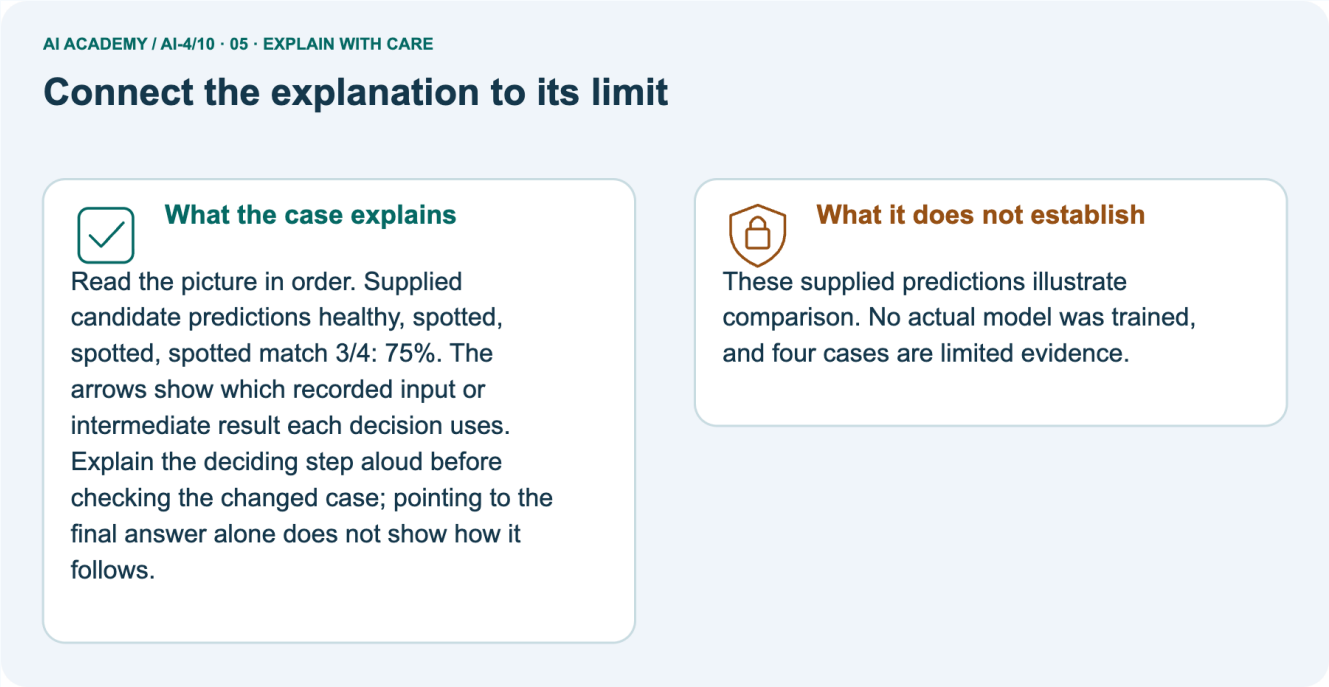
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Changed condition: Change the third candidate **prediction** from spotted to healthy.

Expected result: Candidate becomes correct on all 4 validation cases. Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

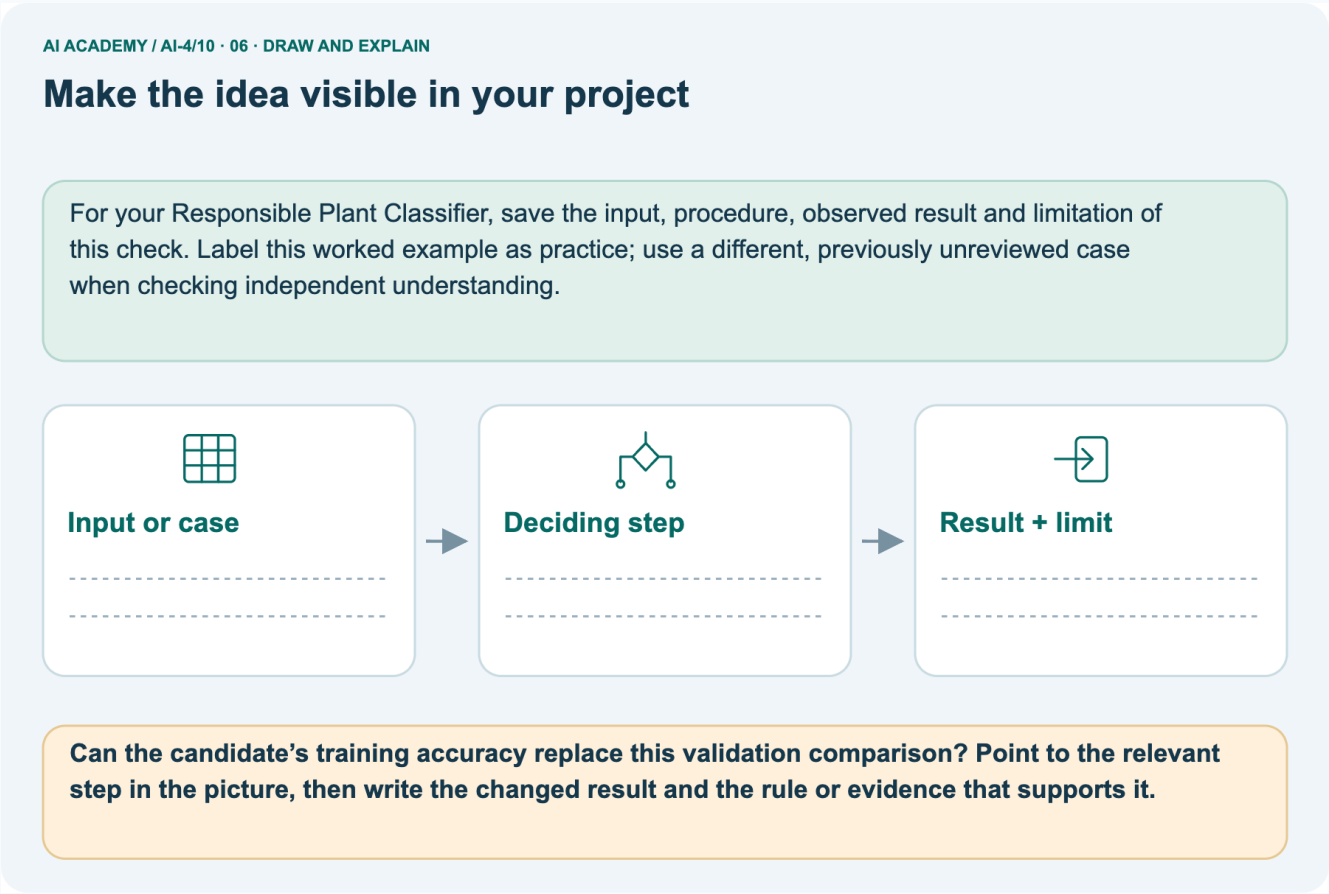
Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: These supplied **prediction**s illustrate comparison.

No actual model was trained, and four cases are limited **evidence**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Baselines First: the case

02. Follow the deciding steps

03. Keep fitting separate from scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.